🤌 XAI: Evaluating Interpretability

FAMAF - UNC

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🧉 Disclaimer ---

This course is based on

Explainable Artificial Intelligence

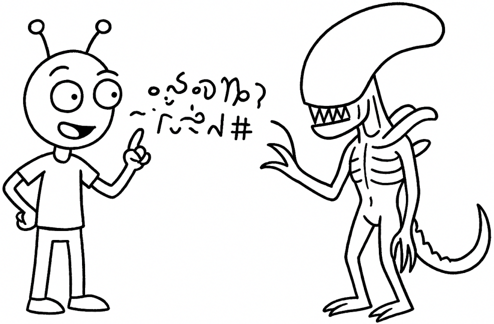
From Simple Predictors to Complex Generative Models

Spring 2023, Harvard University

https://interpretable-ml-class.github.io/

Overview ---

- Evaluating interpretability in the interpretable ML community:
  - Interpretability depends on human experience of the model
  - Disagreement about the best way to measure it
- These papers:
  - Evaluating factors related to interpretability through user studies



🧉 Other Relevant Fields ---

- **Human-Computer Interaction (HCI):**
  - Theories for how people **interact with technology**
- **Psychology:**
  - Theories for how people **process information**
- Both have thought carefully about **experimental design**

🧉 Outline ---

- Research paper: **"Human Evaluation of Models Built for Interpretability"** by Lage et al.
- Research paper: **"Manipulating and Measuring Model Interpretability"** by Poursabzi-Sangdeh et al.
- Discussion

🧉 Paper 1 ---

The Seventh AAAI Conference on Human
Computation and Crowdsourcing (HCOMP-19)

Human Evaluation of Models Built for Interpretability

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Abstract

Recent years have seen a boom in interest in interpretable machine learning systems built on models that can be understood, at least to some degree, by domain experts. However, exactly what kinds of models are truly human-interpretable remains poorly understood. This work advances our understanding of precisely which factors make models inter-

human-simulatable models include human-simulatable regression functions (e.g. Caruana et al., 2015; Ustun and Rudin, 2016), and various logic-based methods (e.g. Wang and Rudin, 2015; Lakkaraju, Bach, and Leskovec, 2016; Singh, Ribeiro, and Guestrin, 2016).

But even types of models designed to be human-simulatable may cease to be so when they are too complex

(Lage et al. 2019)

🧉 Definitions ---

- **Human-simulatability**: The ability of humans to mentally trace through a model's decision-making process and accurately predict its outputs for given inputs, essentially simulating how the model would behave.
- **Decision set**: A logic-based machine learning model consisting of a collection of independent logical rules, where each rule maps a combination of input features to an output prediction.
- **Decision set complexity**: A decision set is more complex when it has more rules, more conditions per rule, or conditions that are more cognitively difficult to evaluate.

🧉 Contributions ---

Research Questions:

- Which types of **decision set complexity** most affect human-simulatability?
- Is relationship between complexity and human-simulatability **context dependent**?

Approach:

- Large scale, carefully controlled user studies

Decision Sets ---

- **Logic-based models** are often considered interpretable
- Many approaches for **learning them from data** (Decision Trees)

faithful and cold or **brave and passive** → **candy or dairy** and **fruit**

sleepy or **patient and obedient** → **spices and grains** or **dairy**

brave and **sleepy or patient or laughing** → **dairy and fruit** or **grains**

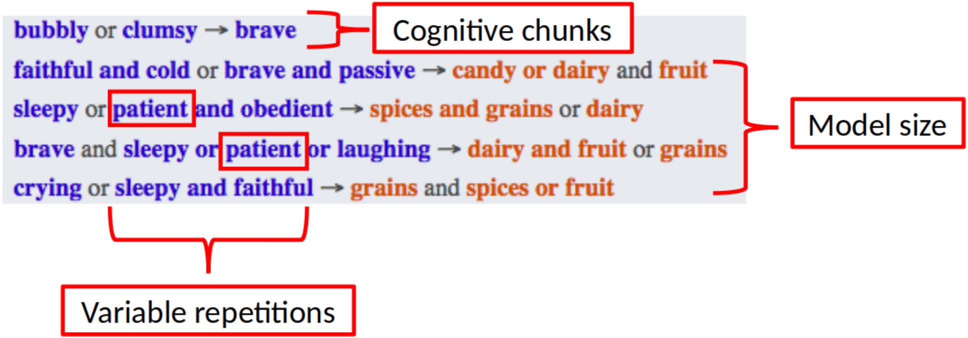
crying or **sleepy and faithful** → **grains** and **spices or fruit**

🧉 Regularizers ---

- **Regularizer (paper context)**: A technique that penalizes certain types of complexity during decision set training (such as number of rules, conditions per rule, or specific types of conditions) to make the models **less complex and more interpretable**.
- What kinds of complexity is it **most urgent to regularize** to learn interpretable models?

🧉 Types of Complexity ---

- **Model size**: Number of rules/lines in the decision set
- **Variable repetitions**: Same variable appearing multiple times
- **Cognitive chunks**: Number of distinct logical conditions



What if we optimized the models with data?

Context: Domains ---

Ingredients:

- **Vegetables:** okra, carrots, spinach
- **Spices:** turmeric, thyme, cinnamon
- **Dairy:** milk, butter, yogurt
- **Fruit:** mango, strawberry, guava
- **Candy:** chocolate, taffy, caramel
- **Grains:** bagel, rice, pasta

Disease Medications:

- **antibiotics:** Aerove, Adenon, Athoxin
- **painkillers:** Poxin, Parola, Pelapin
- **vitamins:** Vipryl, Vyorix, Votasol
- **stimulants:** Silvax, Setoxin, Soderal
- **tranquilizers:** Trasmin, Tydesol, Texopal
- **laxatives:** Lantone, Lezanto, Lexerol

Low Risk: Alien meal recommendation

High Risk: Alien medical prescription

What if we used 2 different real domains?

🧉 Context: Tasks ---

bubbly or **clumsy** → **brave**

faithful and cold or **brave and passive** → **candy or dairy** and **fruit**

sleepy or **patient and obedient** → **spices and grains** or **dairy**

brave and **sleepy or patient or laughing** → **dairy and fruit** or **grains**

crying or **sleepy and faithful** → **grains** and **spices or fruit**

Observations: patient, wearing glasses, lazy

Recommendation: milk, guava

- **Simulation:**
  - What would the model recommend the alien?
- **Verification:**
  - Is *milk and guava* a correct recommendation?
- **Counterfactual:**
  - If *patient* were replaced with *sleepy*, would the correctness of the *milk and guava* recommendation change?

What if we used more realistic tasks?

🧉 Tradeoff Between Control and Generalizability ---

Tradeoff between the ability to *tightly control* the experiment and *running it under realistic conditions* (**generalizability**)

🧉 Procedure ---

- Experiment posted on **Mturk**
- Takes around **20 minutes**
- Participants paid **3 USD**
- Excluded participants who could not complete practice questions
- Total: **50-70** participants **out of 150**

Instructions ➡ 3-6 practice questions ➡ 15-18 test questions ➡ Payment code

🧉 Statistical Analysis: Linear Model ---

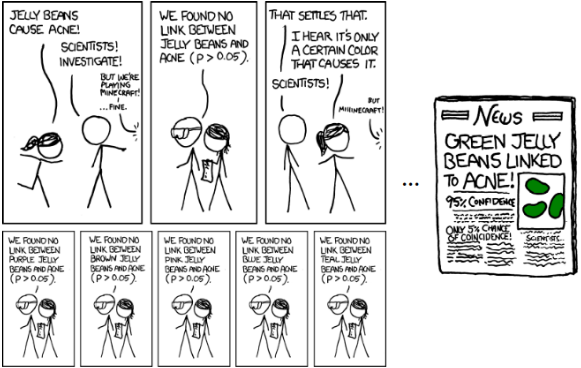
We use a **linear model** for each metric in each experiment:

- Response time
- Accuracy
- Satisfaction

Example – Model Size, Response Time:

- **Step 1:** Fit linear regression to predict response time from number of lines and number of output terms
- **Step 2:** Interpret coefficients as effects of number of lines and number of output terms on response time
  - The coefficient with the **largest magnitude** indicates which factor has the **strongest impact** on response time

🧉 Statistical Analysis: Multiple Hypothesis Testing ---



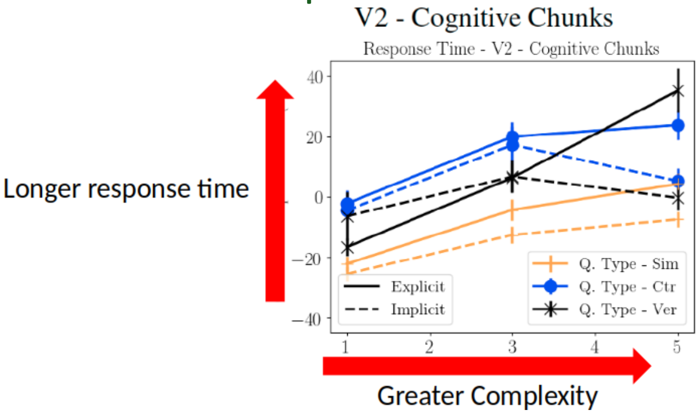
https://xkcd.com/882/

We use a Bonferroni correction

- Instead of **$p < 0.05$**, use: **$p <$ (0.05 / # comparisons)**

🧉 Results: Complexity Increases Response Time ---

Recipe Domain



Key Finding

Greater complexity results in longer response time for all kinds of complexity

Results: Type of Complexity Matters ---

		Response Time			
		Clinical		Recipe	
		Weight	P-Value	Weight	P-Value
Model size	# Lines (V1)	01.17	**<.0001**	01.01	.0032
	# Terms (V1)	02.35	**<.0001**	01.57	.0378
	Ver. (V1)	10.50	**<.0001**	04.11	.1210
	Counter. (V1)	21.00	**<.0001**	13.70	**<.0001**
Cognitive chunks	# Chunks (V2)	06.04	**<.0001**	05.88	**<.0001**
	Implicit (V2)	-13.00	**<.0001**	-07.93	**0.0005**
	Ver. (V2)	16.30	**<.0001**	15.40	**<.0001**
	Counter. (V2)	08.56	.0265	19.90	**<.0001**
Variable repetitions	# Rep. (V3)	01.90	.2470	00.88	.4630
	Ver. (V3)	13.00	.0035	13.70	**<.0001**
	Counter. (V3)	20.30	**<.0001**	16.60	**<.0001**

Significant in one domain

Significant in all domains

Significant in neither domain

Key Finding

Response time for: cognitive chunks > model size > repeated terms

Results: Consistency - Domains, Tasks, Metrics ---

		Response Time			
		Clinical		Recipe	
		Weight	P-Value	Weight	P-Value
Model size	# Lines (V1)	01.17	**<.0001**	01.01	.0032
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For example:
Similar effect sizes, both statistically significant

Key Findings

Results consistent across domains, tasks and the response time and subjective difficulty metrics

🧉 Results: Counterfactuals Are Hard ---

The counterfactual task is much more challenging than simulation!

		Response Time			
		Clinical		Recipe	
		Weight	P-Value	Weight	P-Value
Model size	# Lines (V1)	01.17	<**.0001**	01.01	.0032
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In all experiments, longer response time than simulation

Key Finding

In all experiments, counterfactual tasks require longer response time than simulation tasks

🧉 Discussion: Paper 1 ---

- Consistent guidelines for interpretability
- Simplified tasks to measure interpretability
- Using Mturk workers as a proxy for domain experts

Key Takeaway

Different types of complexity affect *human-simulatability* differently, with cognitive chunks having the strongest effect

🧉 Paper 2 ---

Manipulating and Measuring Model Interpretability

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I] 15 Aug 2021

With machine learning models being increasingly used to aid decision making even in high-stakes domains, there has been a growing interest in developing interpretable models. Although many supposedly interpretable models have been proposed, there have been relatively few experimental studies investigating whether these models achieve their intended effects, such as making people more closely follow a model's predictions when it is beneficial for them to do so or enabling them to detect when a model has made a mistake. We present a sequence of pre-registered experiments ($N = 3,800$) in which we showed participants functionally identical models that varied only in two factors commonly thought to make machine learning models more or less interpretable: the number of features and the transparency of the model (i.e., whether the model internals are

(Poursabzi-Sangdeh et al. 2018)

🧉 Motivation ---

- **Interpretability as a latent property** that can be manipulated or measured indirectly
- What are the factors through which it can be **manipulated effectively?**
- Bring **HCI methods** to interpretable ML since interpretability is defined by user experience

🧉 Contributions ---

Research Questions:

- How well can people estimate what a **model will predict**?
- How much do people **trust** a model's predictions?
- How well can people detect when a model has made a **sizable mistake**?

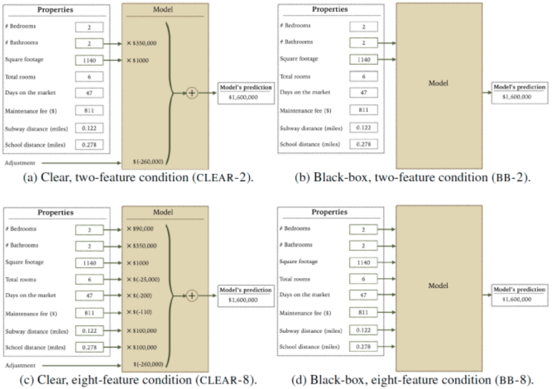
Approach:

- Large-scale, pre-registered **user studies** to answer these questions in the context of **linear regression models**

🧉 Comparison to Paper 1 ---

- Studies **linear regression models** instead of decision sets
- Measures people's ability to make their **own predictions** in addition to forward simulation
- Uses **real-world housing dataset** and models optimized with data

🧉 Ways to Manipulate Interpretability ---



(a) Clear, two-feature condition (CLEAR-2).

(b) Black-box, two-feature condition (BB-2).

(c) Clear, eight-feature condition (CLEAR-8).

(d) Black-box, eight-feature condition (BB-8).

# of Features

Transparency

Procedure ---

- Participants shown:
  - **Training:** 10 apartments
  - **Testing:** 12 apartments (this is the data they use)
- Participants paid **2.5 USD**
- **750-1,250** participants per experiment

Each Trial:

Forward simulate model's prediction → View model's true prediction → Make own prediction

🧉 Statistical Analysis: Participant Specific Effects ---

- A **repeated measures** experimental design
  - Each participant makes many predictions
- Use a **mixed-effects model** to control for correlations between a participant's responses
  - Assumes a random, participant-specific effect

🧉 Statistical Analysis: Multiple Hypothesis Testing ---

- **Pre-registering hypotheses** corresponds to deciding and publishing which analyses you will run **before collecting data**
- Reduces the probability that effects were discovered by chance

Example

4) Specify exactly which analyses you will conduct to examine the main question/hypothesis."

We will use 2-by-2 ANOVA for statistical analysis of the effect of number of features and model clarity on final deviation from model's prediction and simulation error.

https://aspredicted.org/xy5s6.pdf

Design Choices ---

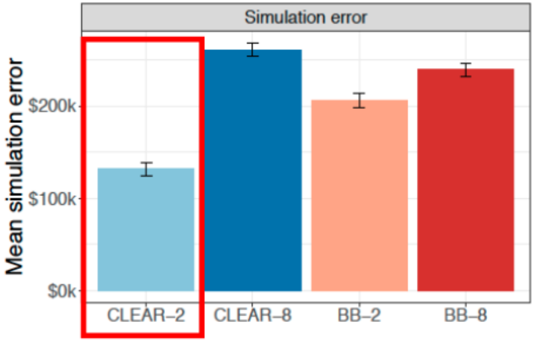
- **Randomized the order** of the first 10 (normal) apartments and **fixed the order** of the last 2 (unusual)
- All participants are shown an **identical set of apartments**
- Each participant completed **a single condition** (between subjects design)

Can introduce bias **Increases variance**

Fix sources of randomness **Randomize as much as possible**

🧉 Results: Simulating Small, Transparent Models ---

Experiment 1: New York City prices

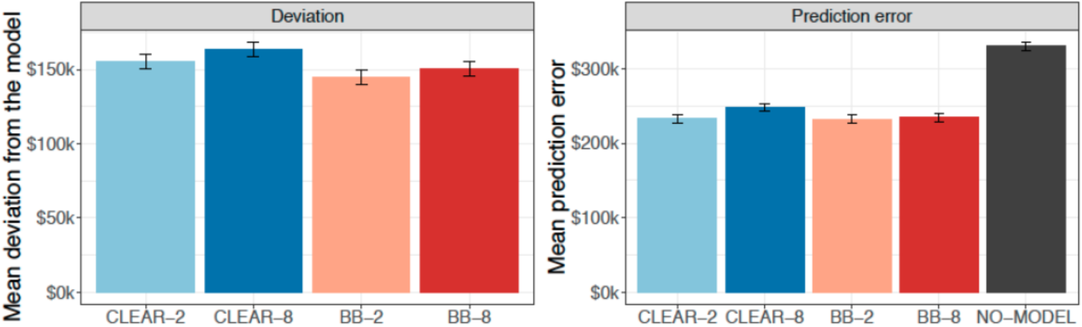


Key Finding

Best simulation accuracy with small, transparent models (CLEAR-2)

🧉 Results: No Difference in Trust or Prediction ---

Experiment 1: New York City prices



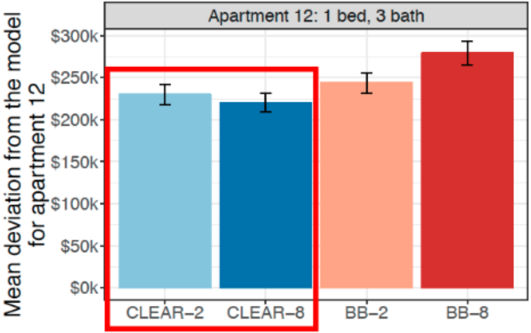
Key Finding

None of the conditions are statistically different for trust or prediction error

🧉 Results: Clear Models Make Mistakes Worse ---

Experiment 1: New York City prices

Deviation
Higher is better



Surprising Finding

Participants deviate *less* from the bad prediction with clear models (CLEAR-2, CLEAR-8)

🧉 Additional Experiments ---

- Scaled down prices to better reflect national average
  - **Same results**
- Better trust metrics
  - **No significant difference in trust between models**
- Attention check for unusual features
  - **People catch more errors** when explicitly asked to look for unusual inputs

🧉 Discussion: Paper 2 ---

Key findings from the experiments:

- Highlighting weird inputs helps catch errors
- Having people predict before seeing the model helped catch errors
- **Transparency actually makes people worse at catching errors**

Counterintuitive Result

More interpretable models (transparent, fewer features) led to worse error detection on unusual cases

🧉 General Discussion ---

Comparing Both Papers:

- **Paper 1:** Decision sets, controlled synthetic domains, cognitive complexity matters
- **Paper 2:** Linear models, real-world data, transparency can backfire

Key Lessons:

- Interpretability is multifaceted and context-dependent
- User studies are essential for evaluating interpretability
- Simplified models $\neq$ always better decision-making
- Different tasks require different evaluation metrics

Thank You!

🧉 References --- |

Lage, Isaac, Emily Chen, Jeffrey He, Menaka Narayanan, Been Kim, Samuel J Gershman, and Finale Doshi-Velez. 2019. "Human Evaluation of Models Built for Interpretability." In *Proceedings of the Aaai Conference on Human Computation and Crowdsourcing*, 7:59–67.

Poursabzi-Sangdeh, F, DG Goldstein, JM Hofman, JW Vaughan, and H Wallach. 2018. "Manipulating and Measuring Model Interpretability. ArXiv."